

SPCE 5065

MIDTERM

You are allowed any course notes, the textbooks by Tribble and/or Pisacane, any material on the Canvas page, and a calculator.

I completed this assignment using only authorized references

Signed: _____

Printed: _____

Problem 1: _____/10

Problem 2: _____/10

Problem 3: _____/15

Problem 4: _____/15

Problem 5: _____/10

Problem 6: _____/20

Problem 7: _____/20

Total: _____

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1. (10 pts.) True/False

- a. _____ Drag reduces an orbit's total and kinetic energy.
- b. _____ Newton's second law states that $\sum \vec{F} = m\vec{a}$, which is directly applicable to rockets.
- c. _____ The coldest layer in the atmosphere is the mesosphere.
- d. _____ When the frequency of galactic cosmic ray events is low, the frequency of extreme solar events is high.
- e. _____ Drag acting on a satellite in LEO reduces its altitude linearly.
- f. _____ Atomic oxygen is the main constituent in the heterosphere during solar min.
- g. _____ Astronauts tend to eat less in free-fall.
- h. _____ Earth's magnetic field axis is constantly moving.
- i. _____ Our current sun cycle (25) is closely matched predications.
- j. _____ GEO satellites do not need protection from the space environment because there is no atmosphere there.

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2. (10 pts.)

I. Which of the effects are *not* techniques to mitigate the neutral environment effects
(*select all that apply*)

- a. Choose materials resistant to atomic oxygen erosion
- b. Shield sensitive surfaces
- c. Bias the spacecraft with a positive voltage
- d. Use coatings on sensitive surfaces
- e. Choose orbits where the most objects are located

II. A mission designer must choose *one* improvement for a CubeSat expected to operate for five years at 500 km altitude. Which option would provide the greatest increase in mission lifetime?

- a. Reduce the spacecraft's frontal area by 40%.
- b. Replace the flight computer with a radiation-hardened processor.
- c. Increase solar array area by 30%.
- d. Add an additional communications antenna.
- e. Increase battery capacity by 50%.

III. Solar ultraviolet radiation alters which of the following, which results in temperature changes on the spacecraft (choose the *best* answer)?

- a. strength to flexibility ratio
- b. reflectivity to transmissivity ratio
- c. absorptivity to emissivity ratio
- d. stress to strain ratio
- e. decomposition to desorption ratio

IV. A Mars transit vehicle is forced to reduce mass by eliminating *one* of the following:

- psychological screening
- simulator training
- exercise equipment
- habitable volume
- medical diagnostic equipment

Which reduction is most likely to produce the greatest increase in overall mission risk?

- a. Psychological screening
- b. Simulator training
- c. Exercise equipment
- d. Habitable volume
- d. Medical diagnostic equipment

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V. What is the primary *cause* of solar flares?

- a. Nuclear fusion in the Sun's core
- b. Magnetic field reconnection
- c. Cosmic ray interactions
- d. Sunspot cooling
- e. Coronal mass ejections

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3. (15 pts) A four-person Mars mission has limited mass available. The project manager proposes eliminating one of the following to reduce launch mass:

- astronaut psychological screening,
- long-duration team training,
- simulator training.

Which would you recommend keeping and why?

Discuss at least **three** ways your choice contributes to mission success, using examples from astronaut selection, stress and coping, group dynamics, or previous space missions.

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4. (15 pts) Your spacecraft has just entered safe mode. Telemetry shows:

- the spacecraft is healthy
- battery voltage is nominal
- multiple computer resets occurred over the South Atlantic
- NOAA issued a G3 geomagnetic storm warning the previous day

Explain:

- a. the most likely cause
- b. what additional data you would request
- c. the immediate operational actions you would take
- d. the long-term design improvements (for future mission) you would recommend

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5. (10 pts.) Initially the Starlink communication satellites were intended to be in LEO at an altitude of approximately 1100 km. Space X later decided to place the satellites at a lower (550 km) altitude due to concerns about the space environment. What might three of these concerns be?

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6. (20 pts.)

a. Given the same Starlink satellite (at an altitude to 550 km), estimate the maximum erosion depth of a Mylar cover oriented on the RAM side during high solar activity over its five year mission life. If the thickness of the mylar cover is 300 μm , will this be a problem? Clearly state your assumptions. Assume an atomic oxygen density of 1×10^8 atoms/ cm^3 .

b. Estimate the decay in altitude the satellite can expect over the five year mission life if no station-keeping is performed. Assume a ballistic coefficient of 103 kg/m^2 . Use an average altitude of 350 km for your density calculations and assume an average R of 6728 km for this problem.

$$\frac{dR}{dt} = -\frac{\rho}{BC} \sqrt{\mu R}$$

Where: $\rho = 1.02 \times 10^7 x^{-7.172} \frac{\text{kg}}{\text{m}^3}$

c. If the thickness of the mylar cover is 50 μm and the deorbit altitude is 150 km, do you expect drag or mylar erosion will be the biggest concern for this mission?

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7. (20 pts.) You are the lead systems engineer for a 12U CubeSat (a 1U CubeSat is 10x10x10 cm) that must remain operational for five years in a 500 km circular orbit.

You may improve only **one** of the following:

- reduce frontal area by 40%,
- increase mass by 50%,
- reduce drag coefficient from 2.2 to 1.5.

Recommend one option and justify your decision using concepts from the neutral environment, ballistic coefficient, solar cycle variability, and orbital perturbations. Discuss any assumptions and tradeoffs.